Bladder Cancer & Diversion

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 129	Tumours of the Urinary Bladder
Ch 130	Management Strategies for Non–Muscle-Invasive Bladder Cancer
Ch 131	Management of Muscle-Invasive and Metastatic Bladder Cancer
Ch 132	Surgical Management of Bladder Cancer: Transurethral, Open and Robotic
Ch 133	Use of Intestinal Segments in Urinary Diversion
Ch 134	Continent Urinary Diversion
Ch 135	Minimally Invasive Urinary Diversion
Ch 124	Bladder Surgery for Benign Disease
★	Putting it together

129 Tumours of the Urinary Bladder

1 · EPIDEMIOLOGY (HEADLINE)

- It is the **fourth most prevalent** cancer and the **costliest per patient over a lifetime**.
- The median age at diagnosis is **73**, and the **male:female ratio is about 3:1**.
- **Women present later and do worse stage for stage**, as haematuria is mistaken for infection.

2 · RISK FACTORS

The exposure predicts the histology.

P2 *CA bladder, etiology, newer investigations and treatment — risk-factor table (occupation, arsenic, Lynch/MSH2, obesity, diet)*

Not in the paper answer:

- **Cyclophosphamide** is the only proven chemotherapeutic carcinogen, via **phosphoramide mustard** (acrolein causes haemorrhagic cystitis, not cancer). Risk is ~4.5× above 20 g, with latency peaking at 10–14 years.
- **Radiation**: external beam radiotherapy for cervical cancer raises risk 2–4×, and for prostate cancer 1.5–2×.
- **Exstrophy** carries a malignancy risk of 3.3–7.5%, and **>90% is adenocarcinoma**.
- **Smoking** causes **30–40%** of urothelial carcinoma (UC) and raises risk **2–3×**.
- **N-acetyltransferase 2 (NAT2) slow acetylators** carry the highest risk, because acetylation detoxifies aromatic amines. Risk falls to near that of never-smokers **>15 years** after quitting.
- **Schistosoma haematobium** causes chronic inflammation and fibrosis, which leads to **squamous cell carcinoma (SCC)**. Long-term catheters in spinal cord injury and stones also cause SCC.
- Germline variants (MSH2, BRCA1/2) occur in 13–24% of UC, so test selected high-grade patients.

3 · PATHOLOGY

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; FGFR3 mutant, TP53/RB wild type	None; resect
Nephrogenic adenoma	Tubule-like glands after injury	None; re-resect if it recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant**, so no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low adenocarcinoma risk, so resect, treat cause and survey

Flat (precursor) lesions

Loss of polarity means dysplasia; loss of umbrella cells means carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	May precede low-grade papillary tumour (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS, so investigate
Atypia of unknown significance	Atypia disproportionate to inflammation	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** progress to CIS if there was prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

UC grade (2004 WHO) – noninvasive spectrum

Non-muscle-invasive bladder cancer (NMIBC) makes up about **70%** of UC and includes Ta, CIS and T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, PI3K, RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20% at 5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

About 80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than their stage.**

Variant	Hallmark	Chemosensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification	No	**Early radical cystectomy (RC) even for NMIBC**
Sarcomatoid (0.3–0.6%)	Large infiltrative mass	Poor	Immediate RC
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse is common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests and inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence and progression
Glandular (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC	2–5% in West; majority where schistosomiasis is endemic	**Secondary is commoner than primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; from the allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic syndromes	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude a primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; 5-fluorouracil–cisplatin by analogy	**En bloc dome with urachal ligament and umbilicus**

KEY POINT

Small cell is systemic disease: primary cystectomy gives poor disease-free survival that adjuvant chemotherapy does not rescue, whereas neoadjuvant chemotherapy markedly improves survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation, showing that inflammation drives it.

Urachal carcinoma (2016 WHO criteria): the epicentre is in the dome or anterior wall, there is no widespread cystitis glandularis, and a secondary source is excluded. TNM staging is unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · MOLECULAR BIOLOGY

	Papillary pathway	Flat pathway
Origin and product	Hyperplasia gives low-grade papillary tumour	Dysplasia gives CIS, then high-grade invasive tumour

	Papillary pathway	Flat pathway
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% of MIBC), chromosome 9 deletion, PIK3CA	**TP53** (up to 76% of MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3**, because they act in the same pathway.

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3 and FOXA1
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- The **consensus classification (2020) has six classes**. Luminal papillary has the best survival, **basal/squamous (35%) is the commonest**, and neuroendocrine-like (3%) is the worst. Luminal-infiltrated tumours respond poorly to cisplatin NAC.

5 · CLINICAL PRESENTATION AND EVALUATION

About one in five visible haematurias hides a cancer.

- **Painless gross haematuria occurs in 85%**. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even on anticoagulants. Check prostate-specific antigen, because ~10% of recurrent gross haematuria is prostate cancer.
- After a negative work-up, repeat it at 3–5 years only if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (AUA 2012); ultrasound may replace it
Cytology/markers	Reasonable	Avoid routinely

The dipstick has sensitivity 91–100% but specificity only ~65%, so **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy with biopsy is the gold standard; flexible equals rigid. White light misses small tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) makes porphyrins build up in tumour, which fluoresce red under blue light	No dye; 415 and 540 nm light is absorbed by haemoglobin, so vessels stand out
Benefit	Finds more Ta and CIS; CIS is seen only on BLC in ~¼	Better CIS detection; lower 12-month recurrence in low risk
Status	Adjunct; only HAL is FDA approved	Adjunct

Urine cytology and biomarkers

P1 P3 *Urine cytology; Asymptomatic haematuria and urinary biomarkers — good for high grade only; Paris System (HGUC N:C ≥0.7); NMP22, BTA, FISH, ImmunoCyt; none replaces cystoscopy*

ALGORITHM

1. Cytology shows high-grade cells but cystoscopy is negative.
2. Take random bladder biopsies and prostatic urethral biopsies.
3. If these are negative, take selective upper-tract washings for cytology, with or without ureteroscopy.

6 · STAGING AND RISK GROUPS

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half of muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **The 2017 change**: regional nodes without distant metastasis are now stage III (previously IV). MCQ Common iliac nodes are N3, not M1a.
- **T1a/T1b substaging** is recommended, because ~50% of T1 are upstaged to ≥T2 at cystectomy.
- **Prostatic urethra**: mucosal CIS is not adverse in itself but escapes intravesical therapy. Only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · DEFINITION AND NATURAL HISTORY

- **Non-muscle-invasive bladder cancer (NMIBC)** is urothelial carcinoma that has not invaded the detrusor: Ta, T1 and carcinoma in situ (CIS). It makes up **70–80%** of new bladder cancers.
- Recurrence occurs in **40–60%**; progression ranges from ~6% in low-grade (LG) Ta to 17% in high-grade (HG) T1.
- **Grade, not stage, is the strongest predictor of progression**, and CIS comes second. Primary CIS progresses in **>50%**.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**.

KEY POINT

NMIBC is two diseases. LG disease is a **recurrence** problem; HG disease is a **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, and cystectomy is for progression that cannot be controlled.

2 · RISK STRATIFICATION

*Risk is **dynamic**, so reassess it at every recurrence.*

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single postoperative chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction of chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG and maintenance to **3 yr**. If BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene or gemcitabine–docetaxel

EORTC and CUETO scoring models

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC **overestimates** risk, because its patients had little maintenance BCG. CUETO, derived in BCG-treated patients, fits better. MCQ Both use **WHO 1973** grading.

3 · ENDOSCOPIC MANAGEMENT

TURBT: principles

- Transurethral resection of bladder tumour (TURBT) quality depends on **complete resection** and **detrusor in the specimen**; missing muscle means higher mortality.
- Image the upper tracts at diagnosis. Upper tract tumours occur in **<5%**, more often with trigonal tumour, CIS and high risk.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria is **T3a**. For HG, strongly consider partial cystectomy or RC.
- **En bloc resection** gives fewer perforations, more detrusor and better staging, but the **same recurrence**.

Repeat (second-look) TURBT

P4 *Re-look (second-look) TUR bladder tumour — indications (all T1, incomplete, no muscle; consider HG Ta) Within 6 weeks; residual ≥40%, HG T1 upstaged 25–30% (~50% without muscle); residual T1 means 80% progression*

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Random biopsies are **not** needed in low risk; consider them with positive cytology or high risk. Biopsy the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences can be fulgurated in the office, but this needs a **prior LG diagnosis and negative cytology**. **Active surveillance** also suits selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; needs intravesical instillation	415 and 540 nm light is absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary tumours	More papillary tumours and CIS

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Weakness	Cost; no 3-yr recurrence benefit in a pragmatic trial	False positives ~60%

4 · SINGLE POST-OPERATIVE INSTILLATION

P3 *Immediate post TURBT intravesical Therapy for TCC — anti-implantation; within 6 h (useless at 24 h); MMC, epirubicin, gemcitabine 2 g (SWOG S0337); best in solitary LG; withhold if perforation; never BCG*

- Give mitomycin C (MMC) within **6 h** (level 1 evidence). After perforation it risks chemical cystitis, contracture and rare deaths.

5 · INTRAVESICAL CHEMOTHERAPY

P4 *Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy — MMC optimisation (Au 2001); Gem/Doce regimen; device-assisted; BCG mechanism, schedule, toxicity*

Not in the paper answer:

- Chemotherapy reduces **recurrence**, with **no proven effect on progression**. In intermediate risk, MMC or gemcitabine equals BCG. Give a 6-week induction, then monthly for 1 yr.
- A quinolone for 24–48 h after each BCG dose reduces cystitis without loss of efficacy.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	FDA approved for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- **Gemcitabine–docetaxel (Gem/Doce)** is increasingly used in BCG-naive HG disease.

6 · BACILLE CALMETTE-GUÉRIN (BCG)

BCG prevents progression, but only with maintenance.

Mechanism

BCG binds fibronectin, releasing cytokines that recruit immune cells and produce a **T-helper 1 (Th1)** cytotoxic response. No strain is superior.

Administration

ALGORITHM

1. Wait **2–4 weeks** after TURBT (re-epithelialisation).
2. Check urinalysis before each dose to exclude infection and gross bleeding.
3. Instil 1 vial in 50 mL saline by catheter and retain it for 1–2 h, restricting fluids beforehand.
4. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection; liver disease (isoniazid cannot then be given for sepsis); previous tuberculosis; anti-TNF drugs; poor performance status; advanced age

- Most absolute contraindications mean a **breached urothelium, allowing intravasation**. *Transplant and steroid patients may receive it, with weaker response.

Efficacy

- BCG is the **only intravesical agent that reduces progression**, and only with **maintenance**.
- In CIS, the initial CR is up to **84%**, and about half are durable.
- A second induction for persistent CIS gives a further 30–50% CR, but beyond 2 courses progression risk rises to 20–50%. **The response at 6 months predicts prognosis.**

Schedule

ALGORITHM

1. **SWOG (Southwest Oncology Group; Lamm 2000)**: give induction weekly ×6.
2. Then give 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**.
3. If high risk, give full dose for 3 yr. If intermediate risk, give 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, lasting <48 h	Symptomatic; consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid, rifampicin and pyridoxine; resume at ½–⅓ dose
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ involvement (liver, lung, kidney)	**Stop BCG**; isoniazid, rifampicin and ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Asymptomatic granulomatous prostatitis needs no treatment.

BCG shortage

ALGORITHM

1. Prioritise high-risk patients.
2. Reduce the dose, especially in maintenance, and split vials.
3. Use chemotherapy: single agent for intermediate, Gem/Doce for high risk. Early RC for the highest risk.

7 · RECURRENCE AFTER BCG

P2 *BCG failure — definitions table: adequate BCG (≥5/6 induction + ≥2/3 maintenance; T1 induction alone), unresponsive (HG within 6 months Ta/T1, 12 months CIS), resistant (reinduce, CR 30–50%), intolerant, delayed relapse*

Not in the paper answer:

- After failure, check the prostatic urethra and upper tracts, because recurrence there develops in about half on long follow-up.

> **KEY POINT**
>
> **Trap:** HG T1 at 3 months after induction alone counts as unresponsive. CIS or HG Ta at 3 months is "resistant", so give more BCG.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-PD-1; 200 mg intravenously every 3 weeks up to 2 yr	41%, then 19%	63%; grade 3–4 toxicity 13% (immune)
Nadofaragene firadenovec*	Adenoviral interferon-α2b gene; intravesical 3-monthly	53%, then 24%	71%; grade 3–4 4%
N-803 + BCG	Interleukin-15 superagonist	71% (any time), then 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	—, then 60% (43% at 2 yr)	84%; no level 1 evidence

*FDA-approved. CR roughly halves by 12 months, and none matches cystectomy.

8 · EARLY RADICAL CYSTECTOMY IN HIGH-RISK NMIBC

- T1 failing BCG at 3 months progresses in **82%**, and CIS in ~50%.
- Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%** (64% if understaged). Even a 12-week delay worsens survival.
- **Patients who progress to muscle invasion under surveillance do worse than those with de novo muscle-invasive disease.**

ALGORITHM

1. **Strongly consider RC** if: BCG-unresponsive HG; persistent HG T1 on re-resection; T1 with CIS, LVI or variant histology; deep lamina propria invasion; diffuse CIS.
2. Also if: diverticular HG tumour; distal ureteric or prostatic ductal involvement; incompletely resectable tumour; patient preference.

9 · SURVEILLANCE

Cystoscopy with cytology is the gold standard. **Tumour at the 3-month cystoscopy** and number at first resection best predict recurrence.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	At 3 months, then annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr, then 6–12-monthly for 2 yr, then annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr**, then 6-monthly for 2 yr, then annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence, using the risk of the **latest resected** tumour.
- Positive cytology with negative cystoscopy needs biopsies and upper-tract work-up.
- Urinary markers (UroVysion FISH is the most specific) do not replace cytology.
- **Extravesical recurrence**: upper tract UC occurs in <1% at low risk and ~10% at high risk. The prostatic urethra is involved in 20–40% of high-risk patients at 10 yr; HG ductal disease needs cystoprostatectomy.

10 · PREVENTION

- **Advise stopping smoking** at every visit, although its effect on recurrence is uncertain.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction with or without maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · NATURAL HISTORY AND HISTOLOGY

MIBC kills through micrometastases, so cure needs systemic treatment too.

- **20–30%** of new bladder cancers are muscle-invasive (MIBC), and **~12%** are node-positive or metastatic.
- **Secondary** MIBC (**~20%**), which has progressed from non-muscle-invasive disease, does worse than **primary** MIBC, with more understaging and less response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy occur within **2 years**, from micrometastases present at diagnosis.
- Urothelial carcinoma (UC) is 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only the use of NAC and radiation.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity, so up-front RC with lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic ACTH, hypercalcaemia	**Chemosensitive**, so NAC with etoposide and cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion (LVI)	If predominant, up-front RC
Mixed UC + variant	Quantify the variant	UC-predominant: NAC; variant-predominant: RC

2 · CLINICAL STAGING

Stage by resection depth, examination under anaesthesia (EUA), computed tomography (CT) with excretory phase, then chest imaging. In cT2, pathology upstages **40–54%**; LVI, variant histology, female sex and secondary MIBC predict upstaging.

TURBT

- Resect all visible tumour if safe, with a separate deep specimen; this is needed for bladder preservation, and pT0 at RC improves survival.

- If high-grade T1 or no muscle, repeat resection within **6 weeks**.
- **Prostatic urethral biopsy** (trigone or bladder-neck tumour, carcinoma in situ (CIS), multifocal disease) is a full-thickness loop from mid-prostate to verumontanum at **5 and 7 o'clock**.
- **Frozen urethral margin at cystoprostatectomy** is mandatory if the prostatic biopsy was positive, and advisable whenever a neobladder is planned; a neobladder is made only if it is negative.

Bimanual EUA

Examine before and after resection with the bladder empty; EUA more often understages than overstages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before resection**, as post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**; hydronephrosis suggests extravesical disease.
- Magnetic resonance imaging stages T best. The **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the muscularis line: intact in 1–3, interrupted in 4, into fat in 5. A score **≥3** is suspicious.
- FDG PET/CT is reserved for equivocal imaging.
- Chest CT is needed for all; bone or brain imaging only with symptoms or raised alkaline phosphatase. Renal function decides NAC and orthotopic diversion.

3 · PATHOLOGICAL STAGING (AJCC 8TH) AND PROGNOSTIC MARKERS

Category (American Joint Committee on Cancer)	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac is **N3, not M1**; T4b N0 is stage IVA. pT4a needs **stromal** invasion, since urethral or ductal CIS does not upstage.
- **Subtypes**: luminal survives best, **basal gains most from NAC**, and p53-like is chemoresistant.
- **Circulating tumour DNA**: over a third stay positive after RC and do badly; in IMvigor010, adjuvant atezolizumab helped only them.

4 · RADICAL CYSTECTOMY: PRINCIPLES AND EXTENT

- **RC with bilateral pelvic lymph node dissection (PLND) is standard for resectable MIBC**, ideally after cisplatin NAC. Delay **>12 weeks** worsens survival.

- **Men**: bladder, perivesical tissue, prostate and seminal vesicles are removed. **Women** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes and anterior vaginal wall.
- **Male organ-sparing** needs no bladder-neck disease and negative prostatic biopsies; incidental prostate cancer is found in ¼–½.
- **Female organ-sparing** needs no base or trigone tumour. Keeping the uterus and anterior vagina supports a neobladder; keep ovaries premenopause unless there is breast or ovarian cancer risk.

> **KEY POINT**
>
> Spare an organ only where it is oncologically silent. The gatekeepers are, in men, the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · PELVIC LYMPH NODE DISSECTION

Nodal status is the strongest predictor of survival.

P4 *Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — templates; German trial and S1011 negative, so standard PLND; extranodal extension*

Not in the paper answer:

- About **25%** are node-positive at RC. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- Standard PLND reaches the common iliac bifurcation; extended reaches the aortic bifurcation. About 10 nodes is the usual cut-off for pelvic recurrence risk.

6 · INTRAOPERATIVE DECISION MAKING

- If **clinically node-positive**, give cisplatin chemotherapy first, then **consolidative RC** in responders.
- **Unexpected nodes**: proceed only if completely resectable within the true pelvis. **Abort RC** if nodes are unresectable, the bladder is fixed to the sidewall or the rectosigmoid is invaded.
- **Ureteric margin** is involved in 6–8% and raises upper-tract recurrence (4–10%, mostly at 2–4 years). Frozen section is not mandated, but send one for gross tumour.
- **Male urethral recurrence** is 1–8% and highest with **prostatic stromal invasion** (~30%), so consider urethrectomy then. A positive pre-op prostatic biopsy does **not** bar a neobladder; the **frozen urethral margin decides**.
- **Female urethra** is involved in 2–12%. Urethrectomy is mandatory only for T4 disease involving urethra or vagina; send a proximal urethral frozen section in every neobladder candidate.

7 · OUTCOMES AFTER RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly from metastases within 2 years.
- **Pelvic recurrence**: low risk (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · NEOADJUVANT THERAPY

Cisplatin NAC gives a level 1 survival gain.

P2 *Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types — cisplatin NAC (ddMVAC 3–6/GC 4), rationale, pT0 ~90% RFS; adjuvant chemo (ABC 2022 +6%), nivolumab, Cairo radiotherapy*

Not in the paper answer:

- About half of patients are cisplatin-ineligible, and they go straight to local therapy.
- Neoadjuvant immunotherapy (IO) phase II trials: **ABACUS** (atezolizumab) complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	Glomerular filtration rate **<60 mL/min**	Creatinine clearance <30 mL/min
ECOG performance status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	NYHA III–IV	NYHA III–IV; multiple high-grade comorbidities

- The NCCN prefers dose-dense methotrexate, vinblastine, doxorubicin and cisplatin (ddMVAC).
- **Neoadjuvant IO** blocks the PD-1/PD-L1 checkpoint and was not yet recommended in the source. †NIAGARA has since shown that perioperative durvalumab with gemcitabine–cisplatin (GC) improves event-free and overall survival (OS).
- Neoadjuvant radiation downstages but gives no survival gain.

9 · ADJUVANT THERAPY

- The target is **pT3–4 or N+**. Post-RC morbidity prevents delivery in ~¼, so neoadjuvant treatment is preferred.
- **Chemotherapy** is not standard, but discuss ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.
- **Nivolumab for 12 months** is given for ≥pT3/N+ after NAC, or if cisplatin-ineligible or declining. IMvigor010 (atezolizumab) was negative.
- **Radiation** may be considered for pT3/N+, where local failure is 20–40%.

10 · BLADDER PRESERVATION

Selected patients keep their bladder with similar survival.

Trimodality therapy (TMT)

P4 *Discuss bladder preservation surgery for TCC of urinary bladder — selection (cT2, no hydronephrosis/CIS; 6–19%), sensitisers, schedules, outcomes, salvage RC 10–30%, follow-up; partial cystectomy*

Not in the paper answer:

- **Dose**: a continuous course to **60–66 Gy** with restaging at 3–4 months is commonest; **55 Gy in 20 fractions** is non-inferior.
- Concurrent chemotherapy beats radiation alone in two randomised trials.

> **KEY POINT**
>
> TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete resection cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months)

Modality	Use	Why it fails
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging; ~25% have occult nodes
Partial cystectomy with PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence, so lifelong follow-up; convert to RC if the frozen margin is positive

11 · METASTATIC UROTHELIAL CARCINOMA

First-line choice hinges on platinum eligibility.

- Bajorin adverse factors are Karnofsky status <80% and visceral metastasis.
- **Cisplatin-eligible**: GC or ddMVAC. **GC equals MVAC** and is less toxic (von der Maase), and dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) with pembrolizumab. †EV-302 later showed EV with pembrolizumab roughly doubled OS vs platinum, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1-positive. Trap: **first-line IO alone is only for platinum-ineligible patients.**
- **Maintenance avelumab** is given if there is no progression after platinum (JAVELIN Bladder 100). IO combined with chemotherapy is not recommended.
- **Second line** is **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); atezolizumab failed (IMvigor211). Platinum can be rechallenged if the last dose was ≥12 months earlier.
- **Antibody–drug conjugates**: **EV** (Nectin-4; neuropathy, rash) beat chemotherapy after platinum and IO (EV-301). **Sacituzumab govitecan** (Trop-2; neutropenia) is third line.
- **Erdafitinib**, an oral FGFR inhibitor, treats FGFR3/FGFR2 alterations after platinum, with 40% response; it causes **hyperphosphataemia**.
- **IO toxicity** includes rash, colitis, hepatitis, pneumonitis, myocarditis and endocrinopathies, which are often **irreversible**.

ALGORITHM

1. If platinum-eligible, give GC or ddMVAC (or gemcitabine–carboplatin, or EV with pembrolizumab).
2. If there is response or stable disease, give avelumab maintenance.
3. On progression, give pembrolizumab, or erdafitinib if FGFR-altered.
4. Then give EV, then sacituzumab govitecan.
5. If platinum-ineligible, give pembrolizumab, then EV.

12 · FOLLOW-UP

- **After RC**: CT with excretory phase and chest CT 6-monthly for 2–3 years, then yearly to 5 (EAU, AUA). Check renal function at each visit and vitamin B12 yearly from year 2.
- Upper-tract recurrence (4–10%) is the commonest **late** one; pelvic and distant recurrences come mostly within 2 years.
- **After bladder preservation**: cystoscopy 3–4-monthly in years 1–2, 6-monthly in years 3–4, then yearly to **10 years**, with CT urogram and chest CT 3–6-monthly for 2 years, then yearly to 5.

13 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3, then RC, vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 is a surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	Cisplatin, methotrexate, vinblastine (CMV) ×3, then RC or radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS hazard ratio (HR) 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free survival 66% vs 56%	ddMVAC better but more toxic
SWOG S1011	Standard vs extended PLND	No difference in disease-free survival (DFS) or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation with or without mitomycin C/5-fluorouracil	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum and IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · FRAMING

- Transurethral resection (TUR) stages every patient and cures low-grade non-muscle-invasive bladder cancer (NMIBC); muscle-invasive disease (MIBC) needs radical cystectomy (RC) and pelvic lymph node dissection (PLND), after cisplatin neoadjuvant chemotherapy (NAC) if eligible.

2 · TRANSURETHRAL RESECTION OF BLADDER TUMOUR (TURBT)

Complete resection with muscle in the specimen makes staging correct.

Pre-operative evaluation

- **Continue aspirin**; hold other anticoagulants and P2Y12 inhibitors by half-life. **Treat active infection before resection**.
- Image the upper tracts by CT urogram, MRI, or ultrasound with retrograde study. Iodinated contrast needs a glomerular filtration rate (GFR) **≥60** mL/min; gadolinium needs GFR **>30**.

Set-up and inspection

ALGORITHM

1. Antibiotic within 1 h (avoid fluoroquinolones).
2. Inspect with 30° and 70° lenses (120° if incomplete).
3. Map tumours; add blue-light or narrow-band imaging if available (more tumours and carcinoma in situ (CIS) found, less recurrence).

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in the specimen** and a **~2 cm** margin of normal tissue where possible.
- Then examine under anaesthesia (EUA) with the bladder empty: a **mass palpable after resection is cT3b**.
- **En bloc resection** keeps orientation and detrusor, and improves T1 staging.
- Maximal resection is therapeutic in NMIBC, but before RC for MIBC it gives no survival gain; residual tumour is only prognostic.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: nerve stimulation makes the leg adduct and the loop perforates	Low distension; bipolar; muscle relaxant (general anaesthesia)
Ureteric orifice	Coagulation causes stricture and hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection or heavy cautery is reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage and observation**	**Laparotomy**, inspect bowel, repair; catheter and drain

> **KEY POINT**
>
> Perforation can follow adequate resection of advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak outside the bladder.

Post-operative intravesical agents

P3 *Immediate post TURBT intravesical Therapy for TCC — rationale, agents, contraindications*

Not in the paper answer:

- Single mitomycin C or epirubicin **within 24 h** gives a ~**13%** absolute fall in recurrence (meta-analysis); gemcitabine (SWOG S0337) cut recurrence by 12% at 4 years. They reduce recurrence, not progression.

- **BCG (bacille Calmette–Guérin) is never given immediately**: wait **2–4 weeks** with no gross haematuria, because of the risk of systemic absorption.

3 · RADICAL CYSTECTOMY: PREPARATION

- **Mark a stoma for every patient** (sitting and standing) and consent for a conduit.
- **Avoid routine mechanical bowel prep** and oral antibiotics; omitting them speeds recovery and cuts wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist), given 30–90 min pre-op and then twice daily, lessens ileus and shortens stay.
- Give an IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- Start **venous thromboembolism (VTE) prophylaxis** (compression and drug) **before induction** and extend it to **4 weeks**: 28 rather than 8 days gives fewer major VTE without extra bleeding. Enoxaparin is standard; apixaban is equal, with better compliance.

4 · OPEN RADICAL CYSTECTOMY: TECHNIQUE

ALGORITHM

1. Lower midline; Retzius space; divide urachus and vas or round ligament.
2. Mobilise caecum and sigmoid; make the **retrosigmoid window** (left ureter).
3. Free and divide ureters with adventitia.
4. Retractor (check vena cava); PLND; staple, clip or seal vesical pedicles.

- **Distal ureteric frozen section** is controversial: a positive margin predicts upper-tract recurrence, but clearing it has no proven benefit.

Male

ALGORITHM

1. Incise peritoneum over seminal vesicles; separate rectum; incise Denonvilliers fascia.
2. Control posterior pedicles; open endopelvic fascia.
3. Ligate dorsal venous complex (DVC); divide urethra.

- **Rectal injury is repaired primarily**, with or without faecal diversion.
- **Urethral frozen section** before a neobladder has a very high negative predictive value. **A positive result rules out orthotopic neobladder**, so warn the patient beforehand.
- **Nerve sparing** leaves few men potent at 1 year, and local recurrence occurs even in pT2. Avoid it in advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing** improves potency and continence, but occult prostate cancer is common. It needs normal DRE and prostate-specific antigen (PSA), a TURP and a negative frozen section.

Female

- The standard is **anterior exenteration**, essential if EUA is positive or cT4a. Discuss **ovarian sparing** if disease is not locally advanced.

ALGORITHM

1. Ligate ovarian vessels; incise posterior fornix; control pedicles; excise urethra.
2. Watertight **clamshell vaginal closure** (keeps girth, loses length).

- **Vagina and urethra sparing for neobladder** needs ≤cT2, an uninvolved bladder neck and a negative urethral frozen section; preserve the striated sphincter.

5 · PELVIC LYMPHADENECTOMY

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	Standard plus common iliac and presacral	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- **≥10 nodes** removed goes with better survival (NMIBC at RC: >20).
- A randomised trial found **extended not superior to limited** for recurrence-free or cancer-specific survival, but its staging value is certain.

6 · PARTIAL CYSTECTOMY

- Indications are a **solitary small MIBC without CIS** and **urachal adenocarcinoma** (en bloc umbilicus, urachus and dome with frozen margins).
- Oncology is as for RC: PLND, NAC if eligible, and consent for RC. Muscle-invasive recurrence needs salvage RC.

ALGORITHM

1. PLND; open the bladder away from tumour.
2. Full-thickness excision with fat and a 1–2 cm frozen-section margin; reimplant ureter if needed.
3. Watertight closure; warm water irrigation (against seeding); drain.
4. Cystogram day 7, then remove catheter.

7 · ROBOTIC RADICAL CYSTECTOMY (RARC)

KEY POINT

No effective salvage exists: urine spillage, poor PLND or a positive soft-tissue margin leads to near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications match open surgery; prior surgery and obesity are relative contraindications. A **15-mm left assistant port** allows stapling and bagging.

ALGORITHM

1. Release sigmoid; develop presacral plane; open cul-de-sac to prostate apex.
2. Open Retzius; control anterior pedicle; free ureters (**never grasp**).
3. **Extended PLND with ureters intact**.
4. Clip ureters, send margin, pass left ureter under sigmoid.
5. Posterior pedicles, DVC, urethra (clip proximally only if neobladder).

- PLND before ureteric division keeps urine output as a resuscitation guide. Bag each packet whole; the obturator nerve is **not** the posterior limit, so go down to the hypogastric vein.

8 · COMPLICATIONS AND POST-OPERATIVE CARE

- Gastrointestinal complications lead, then infection. For **ileus**, use alvimopan, early feeding and opioid-sparing analgesia, with **no routine nasogastric tube**.
- Ureteric devascularisation causes ischaemic stricture (keep adventitia).

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Grossman 2003	NAC and RC vs RC	Median survival 46 vs 77 months	Offer cisplatin NAC
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent
iROC	RARC with intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · SURGICAL ANATOMY

A diversion depends on its pedicle's blood supply.

- **Stomach** is supplied by the right gastroepiploic (gastroduodenal, from hepatic) and left gastroepiploic (from splenic) along the greater curve. The graft is antrum or a fundal wedge, which must **never reach the pylorus or lesser curve**.
- **Small bowel** is ~22 ft: the proximal 2/5 is jejunum (wider, single arcades) and the distal 3/5 ileum (multiple arcades). Bowel >**8 cm** from a straight vessel may not survive, so clear ≤8 cm of mesentery at a segment end.
- **Avoid pelvic small bowel after radiotherapy**, including the fixed terminal ileum. A right colic arising from the ileocolic lets the ascending colon reach the pelvis.
- **Colon** is supplied by the superior mesenteric artery (SMA: ileocolic, right colic, middle colic) and inferior mesenteric artery (IMA: left colic, sigmoid and superior haemorrhoidal). The internal iliac gives the middle haemorrhoidal; all link through the **marginal artery of Drummond**.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic and ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid and superior haemorrhoidal

> **KEY POINT**
> Watershed zones get the smallest end branches of two arteries, so they go ischaemic first in hypotension or embolism; site anastomoses to one side.

2 · SEGMENT SELECTION

P2 P1 *Use of intestinal segments in urology / Intestinal segments in urology — segment table (electrolytes, pros, cons, when to choose), NH4Cl mechanism, alkali therapy*

Not in the paper answer:

- Losing **ileum** causes B12, bile-salt and fat malabsorption; losing the **ileocaecal valve** causes bacterial overgrowth, diarrhoea and bicarbonate loss. **B12** liver stores last **3–5 yr**, so give empirical parenteral B12 after 5–7 yr of ileal diversion.
- **Stomach** suits renal failure, short gut and dense adhesions, but causes the **haematuria–dysuria syndrome** (treat with H2 blockers). Use **jejunum** only if nothing else works.

3 · BOWEL PREPARATION AND ENHANCED RECOVERY

P3 *Methods of preoperative bowel preparation — mechanical lowers amount, antibiotics lower concentration; oral regimens and harms; IV prophylaxis most important; ERAS; who still needs full*

prep

Not in the paper answer:

- The descending colon holds **10$^{10–12}$/g** bacteria; oral antibiotics cut this to ~10^2/g. The **IV antibiotic**, given 1–2 h before incision and covering anaerobes, matters most.
- **Alvimopan** (peripheral μ-opioid antagonist) is given as 12 mg 30 min–5 h pre-op, then twice daily, limited to 15 doses for inpatients.
- **Enhanced recovery after surgery (ERAS)** uses **no routine bowel prep**; postoperative ileus is the commonest cystectomy complication.
- Salvage or benign cystectomy (radiation, fistula, neurogenic) still needs full mechanical and antibiotic prep.

4 · INTESTINAL ANASTOMOSES

1. **Get exposure** and wall off the field; check the **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage** with non-crushing clamps, and irrigate the segment.
3. Join **serosa to serosa, inverted** and watertight, with **no strangulating sutures** and the **mesentery aligned**.

- Breakdown follows ischaemia, faecal sepsis, a **drain on the anastomosis** or irradiated bowel (most lethal). Sutured and stapled are equal. Leak presents late (2nd week); a neobladder–vaginal fistula causes continuous incontinence.
- **Obstruction** occurs in ~10% with ileum or stomach and ~5% with colon, mostly from adhesions. Prevent it with unirradiated bowel, closed mesenteric apertures, omentum and **reperitonealisation**.
- **Ogilvie syndrome** is a dilated caecum; exclude obstruction with a water-soluble enema. If the caecum is **≥12 cm** or white count rises, give neostigmine or decompress by colonoscopy; operate for ischaemia or peritonitis.

5 · URETEROINTESTINAL ANASTOMOSES

- Mobilise minimally, keep the periadventitia, spatulate, and sew mucosa to mucosa. **Bring the bowel to the ureter**, retroperitonealise, and use soft stents (they cut leak from 3–5% to near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel made from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Ureters joined together (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

> **KEY POINT**
>
> Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation, so a stented refluxing anastomosis is the default; reflux also allows a loopogram.

- **Stricture** follows ischaemia, leak, radiation or infection, so surveillance is lifelong. The classic site is the **left ureter under the IMA**.
- Open reanastomosis is most durable; endoincision works in ~50–60% at 2 yr. Endourology fails more if the stricture appears <1 yr post-op, is **≥1.5 cm**, or is left-sided.

6 · CONDUIT DIVERSION

- The **ileal conduit** is simplest, with fewest early complications; avoid it in short bowel, inflammatory small-bowel disease and irradiated ileum.

ALGORITHM

1. Take **10–15 cm** of ileum, 10–15 cm from the ileocaecal valve, keeping a major arcade.
2. Restore continuity by ileoileostomy; close the mesenteric window; flush the segment.
3. Pass the left ureter behind the sigmoid mesentery; make stented anastomoses.
4. Fix the base retroperitoneally in the right lower quadrant (RLQ); make the stoma.

- Intractable stomal bleeding suggests portal-hypertensive varices, treated by TIPS or embolisation.
- A **jejunal conduit** (15–25 cm from Treitz, left upper quadrant stoma) is used only if ileum and colon are unusable.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Avoid colon conduits in inflammatory bowel disease and chronic diarrhoea. An **ileal vesicostomy** suits neurogenic bladder and is reversible.
- Remove stents at days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

A gastric conduit causes no acidosis (0%).

7 · STOMAS

- Mark sitting and supine; go **through the rectus** (lateral stomas herniate); stay ≥5 cm from the incision and avoid creases, scars and irradiated skin.
- **Keep the subcutaneous fat**, cut the fascia in a cross, and make the aperture admit two fingers.
- A **nipple (rosebud)** stoma suits an appliance (2–3 cm spout, less stenosis); a **flush** stoma suits a continent catheterisable channel.

	End stoma	Loop (Turnbull)
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Parastomal hernia occurs in 2–6.6% of ileal conduits. Stomal stenosis occurs in 20–24% of ileal and 10–20% of colon conduits, and in >50% of paediatric catheterisable stomas.

8 · RENAL FUNCTION

- Renal deterioration after conduits is 10–74% (ileal conduit 18%), and ~6% die of renal failure. After ureterosigmoidostomy it is the leading cause of death beyond 15 yr.

- Continent diversions demand more of the kidney than conduits. **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Test	Threshold for continent diversion
Creatinine and urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- **Bowel invalidates** urine osmolality and pH and serum urea and creatinine, because they are reabsorbed.

9 · METABOLIC COMPLICATIONS

Absorption depends on the **segment, surface area, dwell time, solute concentration, renal function and pH**.

Segment	Na	K	Cl	pH	Associated	Treatment
Stomach	—	Low	Low	High	Raised aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	Low	High	Low	Low	Raised renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide
Ileum/colon	—	Low	High	Low	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

> **KEY POINT**
> Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na and Cl, high K); ileum and colon absorb NH4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis** is usually trivial, but severe with renal failure or dehydration. Above a **gastrin** of **120 ng/L**, small gastrin rises cause large bicarbonate rises. Treat stepwise: rehydration and H2 blocker, arginine HCl if life-threatening, omeprazole, then segment removal.
- In the **jejunal syndrome**, NaCl loss causes hypovolaemia and activates renin–aldosterone, so the kidney keeps losing K-rich urine. It is worse with proximal segments, can be fatal, and needs long-term oral salt and thiazide.
- **Hyperchloraemic acidosis** is worst with **ureterosigmoidostomy** and colonic pouches; $NH4^+$ also enters via K^+ channels. Mucosal villous atrophy occurs, but transporters persist, so the risk persists.
- Alkali options are sodium bicarbonate, citrate solutions, or potassium citrate when K is needed. **Chlorpromazine 25–50 mg tds** or **nicotinic acid 400 mg 3–4×/day** inhibit cyclic AMP; they limit, but do not correct, acidosis.
- **Hypokalaemia** is commoner with colon. **Trap**: correcting acidosis alone causes profound hypokalaemia and flaccid paralysis, so give K with bicarbonate. Mg and Ca are also low.
- **Altered sensorium** is usually **ammonia** (liver dysfunction plus ureterosigmoidostomy). Treat by draining the diversion, oral neomycin, protein restriction, IV arginine glutamate and lactulose.
- **Drugs are reabsorbed**, notably **phenytoin, some antibiotics and methotrexate**, so **drain continent pouches during chemotherapy**.
- **Osteomalacia** is commonest with ureterosigmoidostomy, from acidosis, vitamin D resistance and renal Ca loss. Alkaline phosphatase is raised. Correcting a base deficit >2.5 mEq/L prevents bone loss; treat with alkali and calcium, then active vitamin D.
- **Infection**: ~3/4 of conduit urines are bacteriuric; **treat pure Proteus or Pseudomonas** and observe asymptomatic mixed cultures.

- **Stones** (struvite, Ca phosphate) follow acidosis, urease organisms, **staples** and mucus: colon conduit 3–4%, ileal conduit 10–12%, continent caecal reservoir ~20%.

Cancer

P4 *Secondary tumours after urinary diversion — Kälble risk table by diversion, surveillance, pathogenesis, adenocarcinoma at ureterocolic junction*

Not in the paper answer:

- After **ureterosigmoidostomy**, cancer develops in 6–29% after **10–20 yr**. Defunctioned anastomoses still develop cancer, so **excise them, do not ligate**.

10 · NEUROMECHANICAL ASPECTS

- Detubularising bowel into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure; this is the basis of reservoirs.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · DEFINITION AND BASIC PRINCIPLES

P4 **P2** *Continent urinary diversion: orthotopic vs non-orthotopic / Principles of continence in continent diversion — definition, three principles (reliable outlet, low pressure by detubularisation + sphere/Laplace, capacity ≥500 mL; 44 cm ileum <200 to 500–600 mL in a year), ileum stretches more than colon*

Not in the paper answer:

- Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous suits pre-existing incontinence or urethral disease.

> **KEY POINT**
>
> Continence is a pressure balance: low reservoir pressure (from detubularisation) against adequate outlet resistance (sphincter or efferent valve). If either fails, the patient leaks. Camey I (tubular ileum) failed because its pressure was too high.

2 · PATIENT SELECTION

Selection means excluding a bad urethra, kidneys or hands.

P4 *Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion — eligibility 80–90% men / ~75% women, counsel all options (AUA/EAU), absolute and relative contraindications table, renal thresholds, ~10% intraoperative change, age/obesity not bars*

Not in the paper answer:

- **Locally advanced or node-positive (N+) disease found at surgery is not a contraindication**: pelvic recurrence is ~7% (13% if extravesical or N+) and rarely disables the neobladder. The exception is a grossly positive margin at pubis, rectum or side wall.
- eGFR 40–59 is safe, as decline is no faster than with a conduit.
- A short or fatty mesentery is the commonest technical reason to abandon orthotopic diversion.

ALGORITHM

1. Estimated glomerular filtration rate (eGFR) >40.
2. Intact external sphincter and adequate bowel.
3. Not frail; no severe liver, cardiac or psychological disease.
4. Men: no distal prostatic urethral cancer. Women: no palpable anterior vaginal tumour or cervical invasion.
5. If all are met and the patient will self-catheterise, offer an orthotopic neobladder.

- Mark a conduit site in everyone, through mid-rectus and away from creases, umbilicus and scars.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+, so give neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside the verumontanum — optional	Examination under anaesthesia (EUA); bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; the American Urological Association (AUA) requires a negative margin)	
Bladder carcinoma in situ (CIS)	Not predictive	
Skip lesions	—	None: a negative bladder neck means a negative urethra

- Risk rises with depth: none < CIS/mucosal/ductal < stromal. **The neobladder itself is protective** (hazard ratio (HR) 0.44).
- Prostatic involvement needs urethral surveillance (wash cytology, urethroscopy).
- In women, keep the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.

Patient-related factors

- **Renal function** is the key contraindication. Proceed if creatinine **<1.7–2.2 mg/dL** or eGFR **>35–40 mL/min**, with hydration and oral bicarbonate and no nephrotoxins.
- **Self-catheterisation** is needed by 10–20% of men and ~30% of women with a neobladder, and by all with cutaneous pouches. The need may appear years later, so all must be willing; multiple sclerosis, quadriplegia or frailty favour a conduit.
- **Sphincter weakness** is managed with an artificial urinary sphincter (AUS) in men, and Burch or sling in women (who will likely catheterise).
- **Prior radiation** is acceptable without incontinence, stricture, fistula or severe bowel damage; the final decision is made intraoperatively.
- After **prior prostatectomy**, pre-op continence predicts outcome. After **prior bowel resection** with short bowel, use a sigmoid neobladder.

3 · PRESERVING CONTINENCE AT CYSTECTOMY

The **rhabdosphincter** (pudendal nerve) is critical. It lies **anterior and lateral** to the urethra in both sexes.

Men

- Never dissect posteriorly beyond the apex. Incise endopelvic fascia and puboprostatic ligaments minimally; control the dorsal venous complex (DVC) with **absorbable** suture, as clips or staples erode into the anastomosis.

ALGORITHM

1. Open the endopelvic fascia; incise the puboprostatic ligaments slightly.
2. Ligate the DVC close to the apex.
3. Incise the anterior urethra and place sutures.
4. Place posterior sutures through the rectourethralis, then divide the posterior urethra.
5. Tag the sutures until the ureteroenteric anastomosis is done.

- Place sutures **before** full division, as the male urethra retracts. Apical frozen section with tumour means urethrectomy and cutaneous diversion or conduit; atypia or focal CIS means a second distal margin.
- Nerve sparing has a controversial continence benefit; it may help night continence by keeping urethral sensation.

Women

- Sparing uterus and anterior vagina in selected low-stage disease avoids fistula, improves sexual function and reduces retention. A palpable bladder-neck or vaginal tumour needs en bloc urethrectomy and cutaneous diversion.

ALGORITHM

1. Divide the posterior pedicles and develop the plane off the anterior vagina.
2. Use the Foley balloon to mark the bladder–urethra junction.
3. Free the bladder neck and the first 0.5 cm of urethra.
4. Clamp the bladder neck, divide just distal to it, and send a frozen section.

- Leave endopelvic fascia untouched; the bladder neck is not needed for female continence.
- Always interpose an **omental flap** between vagina and pouch, and keep vaginal and urethral suture lines apart.

4 · CHOICE OF BOWEL AND URETERIC REFLUX

Ileum gives lowest pressure; obstruction, not reflux, loses kidneys.

- **Ileum is the default**, being most distensible; its mucosa atrophies, lowering mucus and absorption. Colon has pressure waves; stomach has poor compliance and haematuria–dysuria. Close with absorbable suture, as staples cause stones.
- **Antireflux is unnecessary**: Valsalva raises pressure equally throughout, and every valve can obstruct. A direct (Wallace or Leadbetter) anastomosis strictures in **3–6%**, against 7–29% for Le Duc. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr) and silent.

5 · ORTHOTOPIC NEOBLADDER DESIGNS

P2 P4 *Use of intestinal segments in urology, and orthotopic neobladder — design table (Studer, Hautmann, T pouch, Abol-Enein, Camey II, Hemi-Kock, Padua, Mainz III, sigmoid); Studer steps; Hautmann vs Studer*

Not in the paper answer:

- Hautmann became refluxing in 1997, cutting stenosis from 9.5% to 1%. Studer is the most popular design; the sigmoid neobladder has the worst night continence.

ALGORITHM

1. **Studer:**
2. Isolate ileum ending 15–20 cm proximal to the ileocaecal valve; keep a 40–44 cm reservoir (two 20–22 cm limbs) and an afferent limb.
3. Open the reservoir, oversew the back wall, and fold it the opposite way.
4. Make stented end-to-side ureteroileal anastomoses.
5. Join the dependent point to the urethra.

KEY POINT

Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · CONTINENT CUTANEOUS RESERVOIRS

P2 *Principles of continence in continent urinary diversion; Mitrofanoff principle — nipple vs flap vs imbricated valve, appendix ≥5–6 cm/≥12 Fr, Mitchell extension, Yang–Monti, tapered ileum, Indiana*

Not in the paper answer:

- **Indiana** uses the right colon and ~10 cm of terminal ileum, with a double-imbricated ileocaecal valve and tapered ileum; the whole colon is opened to abolish caecal peristalsis.
- An **umbilical stoma** has less stenosis and is easiest for paraplegics; the appendix takes only 14–16 Fr catheters.

7 · PERIOPERATIVE CARE

- **Enhanced recovery after surgery (ERAS)** means prehabilitation, carbohydrate loading, restrictive fluids, no nasogastric (NG) tube, early diet and opioid-sparing analgesia. **Alvimopan** cut ileus from ~23% to ~7%; ileus is the commonest gastrointestinal complication.
- **Drainage**: single-J stents and a 24 Fr urethral catheter irrigated for mucus; remove the catheter at **3 weeks**. Then use timed voids, lengthened progressively, with pelvic floor exercises.

8 · RESULTS AND COMPLICATIONS

Men tend to leak, women to retain.

Early (<90 days) morbidity is no higher than with an ileal conduit.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; if undrained, percutaneous drain with or without nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or Proteus/Pseudomonas; in febrile urinary tract infection (UTI), look for obstruction, stones, residual; if recurrent, nitrofurantoin or methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	Suspect on CT or ultrasound (US); confirm by pouchogram or antegrade study; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter with or without drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond the first weeks it won't heal, so repair transvaginally or transabdominally, or convert to cutaneous
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; explore if below fascia
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence comes over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2, and nocturnal incontinence persists in **20–50%**. Risks are age >65, colon, diabetes, no nerve sparing and hysterectomy.

- Night leakage reflects loss of the sphincter–detrusor reflex; advise voiding at bedtime and an alarm 2–3 times a night.
- Evaluate persistent leakage after the 12–18-month plateau, excluding fistula and using urodynamics.

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule		No blind retropubic needle passage; use bone anchors or the prepubic route

Retention

- It presents as infection, overflow, palpable pouch or new hydronephrosis. It affects men in 4–10% (~21% by 10 yr) and **women in 20–30%**.
- In women the cause is **mechanical**: the pouch falls back and angulates the pouch–urethral junction. Prevent with omentum behind the pouch, vaginal suspension, sacrocolpopexy and uterine preservation.
- Treat with clean intermittent self-catheterisation (CISC), as drugs are useless; exclude stricture or recurrence.

> **KEY POINT**
> Men fail by **leaking** (sphincter without reflexes), so augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks), so support the pouch.

9 · FOLLOW-UP

- Stricture occurs in the first 4 months, recurrence from 4 months to 3 yr, and stones, late obstruction and urethral or upper-tract tumour after 3 yr.
- Review 4-monthly in year 1, 6-monthly to 3 yr, then annually (examination, metabolic panel, blood count), with CT and chest imaging to 5 yr.
- Check B12 yearly. Do urethral wash cytology in men, and bimanual examination and voided cytology in women. After 5 yr continue lifelong upper-tract US, metabolic panel and blood count.
- No diversion has proven superior quality of life; the neobladder gives better body image and the conduit better urinary function.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 radical cystectomy (RC) patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · CONTEXT

- Robot-assisted radical cystectomy (RARC) equals open oncologically in trials (RAZOR), but with **extracorporeal** diversion (ECUD); intracorporeal diversion (ICUD) is untested.
- In the International Robotic Cystectomy Consortium (IRCC), ICUD rose ≈**11%/year**, driven by the **ileal conduit**, the commonest robotic diversion.

2 · PATIENT SELECTION

A bar to one step is not a bar to all.

Obesity, age, advanced disease and prior surgery no longer exclude.

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function (as for open)

ALGORITHM

1. Pasadena Consensus learning pathway:
2. RARC with extracorporeal diversion.
3. Intracorporeal conduit (≈30–100 cases).
4. Continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · PERIOPERATIVE BUNDLE

Enhanced recovery (ERAS) uses alvimopan, goal-directed fluids, immediate oral diet and opioid-sparing analgesia, giving less ileus and a shorter stay. Caution: a stay **<5 days** about doubles 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, and no oral antibiotics (electrolyte upset, liquid spillage, no benefit)
Thromboprophylaxis	Pneumatic compression and stockings; low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · POSITION AND PORTS

- Use lithotomy and steep Trendelenburg, with Hasson entry after prior surgery.
- **Six to seven ports sit more cephalad** to reach over small bowel and take nodes to the aortic bifurcation.
- A neobladder adds a left stapling port; a suprapubic 12-mm port serves bowel reanastomosis and extends for specimen extraction.

5 · NERVE SPARING

One plexus serves sexual function and neobladder continence.

	Male	Female
Danger points	Presacral and medial common iliac; high pedicle ligation; seminal vesicles and Denonvilliers fascia; prostatic pedicles and apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)

	Male	Female
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra and anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third, so continence survives loss of bladder neck

6 · INTRACORPOREAL ILEAL CONDUIT

ALGORITHM

1. Pass the left ureter under the sigmoid mesentery to the right.
2. Choose 12 cm ileum, 15–20 cm from the ileocaecal valve; place the marionette stitch.
3. Isolate with a 45-mm stapler.
4. Do ureteroileal anastomoses (left first) and stent.
5. Reanastomose bowel, then form the stoma.

- The **marionette stitch** (1-0 silk through the stoma site and conduit end) is held **untied**; it replaces the missing hand and positions the conduit.
- Avoid a redundant segment (See Ch 133), and tag cut ends to prevent malrotation.
- The **enterotomy** is an athermal cold-scissors buttonhole, because thermal slits gave more strictures.
- Single-J stents are used, and the **guidewire comes out only after anchoring**. Use a Bricker (separate end-to-side) or Wallace (Y, one anastomosis) technique.

7 · INTRACORPOREAL NEOBLADDER

The key change is **early neobladder–urethral anastomosis** (Karolinska): dependent ileum is anchored to the urethra first, then the pouch is built in place. Design principles: See Ch 134.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM

1. Urethral anastomosis (Van Velthoven barbed suture).
2. Detubularise distal 40 cm (cold scissors); form the posterior plate.
3. Join ureters to afferent limb or chimney, stented.
4. Close the anterior wall once the 22-Fr catheter is confirmed.

- A **tension-free urethral anastomosis** lacks tactile feedback. It is helped by urethral length, less Trendelenburg, perineal pressure and Penrose traction. Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM

1. Mobilise the right colon.
2. Divide ileum 15 cm from the ileocaecal junction and transverse colon proximal to the middle colic; restore continuity.
3. Detubularise along the anterior taenia and fold.
4. Bring ureters through the posterior plate on double-J; place a caecostomy.
5. Taper ileum over 14 Fr, imbricate the valve, and form an umbilical stoma.

KEY POINT

Capacity comes from detubularised right colon; continence from tapered ileum and the imbricated ileocaecal valve.

8 · POSTOPERATIVE CARE

Avoid overhydration; there is no routine nasogastric tube (Cochrane). Irrigate the neobladder **8-hourly** for mucus.

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks** , after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically at 3–4 weeks
Feeding-tube stents tied to Foley	With the catheter

9 · OUTCOMES

Most problems come from the diversion, and many appear late.

- **Operative time** depends most on diversion type (conduit ≈70 min shorter); ICUD and ECUD take similar time. Proficiency needs ≥**66 RARCs**.
- **Complications** are mostly diversion-related, infection commonest. Two-thirds of reoperations occur after 3 months, so 90-day reporting underestimates them.
- **Parastomal hernia** reaches ≈one-third by 3 years (risks: long operation, fascial defect ≥30 mm, low eGFR).
- ICUD versus ECUD (IRCC) is neutral for high-grade complications and readmission after adjustment, with less transfusion.
- **Reoperation** reaches ≈half by 5 years; predictors are neoadjuvant chemotherapy and any high-grade complication.
- **Uretero-enteric stricture** rises over 5 years; robotic revision equals open.

KEY POINT

Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep, so continence depends on reservoir capacity. Hence detubularisation, nerve and sphincter preservation, a night alarm and limiting night output. Assess at 6–12 months.

- Good ECOG status and retained filling sensation predict continence. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.
- Intracorporeal series report day continence 68–100%, night 55–85%, and CIC up to 11%.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD

Trial	Design	Result	Take-home
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

124 Bladder Surgery for Benign Disease

1 · SURGICAL ANATOMY ESSENTIALS

- The bladder is an extraperitoneal organ (mucosa, lamina propria, detrusor, perivesical fat) derived from the endodermal urogenital sinus.
- **Anteriorly** lies the space of Retzius, the usual access. **Posteriorly**, peritoneum reflects near the prostatorectal junction (men) or vesicouterine space (women), and Denonvilliers fascia separates bladder and prostate from rectum. **Laterally** lie the pelvic sidewall, obturator space, and vas or round ligament.
- The **medial umbilical ligaments** are the obliterated umbilical arteries; the **median umbilical ligament** is the urachus, which can persist as a patent urachus, sinus or cyst.
- The **uterine artery** crosses over the ureter near the internal os, and the **vas** crosses it near the seminal vesicle tip. The distal ureter runs medial to the medial umbilical ligament.
- Superior and inferior vesical arteries form the posterolateral pedicles; lymph drains mainly to external iliac nodes. Nerves are sympathetic T10–L2 and parasympathetic S2–S4; the pudendal nerve (Onuf's nucleus) supplies the sphincter.

KEY POINT

Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety. They decide whether surgery stays extraperitoneal.

KEY POINT

Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley for about **1 week**, drain out at about **48 h** if output is low, and a **cystogram before catheter removal**. With high drain output, drain creatinine above serum means a urine leak.

2 · BLADDER DIVERTICULA

A sac without muscle cannot empty: stasis, infection, stones, cancer.

P2 P3 *Bladder diverticulum / Hutch diverticulum — definition (no muscularis propria), congenital vs acquired table, VCUG/cystoscopy/CT/urodynamics work-up, indications, tumour inside means cystectomy*

Not in the paper answer:

- Voiding cystourethrogram (VCUG) shows hidden diverticula and the outlet obstruction, and decides between a preoperative stent and simultaneous reimplantation.
- A fibrous **pseudocapsule** gives the surgical plane. Most acquired diverticula are asymptomatic and incidental.
- CT urogram shows hydronephrosis, ureteric displacement and stones. Send cytology in all, add flow rate or urodynamics, and obtain a **negative urine culture** before surgery.
- At cystoscopy, enter every sac and note neck size and distance from the orifice; treat stricture or bladder neck contracture before or with the diverticulectomy.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection
Large with residual urine, recurrent urinary tract infection (UTI), stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy with or without reimplantation
Cancer in diverticulum	Partial or radical cystectomy, not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**, and augmentation for neurogenic bladder. Do **endoscopic outlet surgery first**, because later irrigation disrupts fresh suture lines.
- Localise the sac by selective filling or flexible cystoscopic transillumination, and place a ureteral catheter if it lies near the ureter.
- Open extraperitoneal surgery remains the gold standard, even for posterior sacs; laparoscopic and robotic surgery is reasonable in experienced hands.

ALGORITHM

1. **Open intravesical diverticulectomy**
2. Fill the bladder with about 200 mL via an 18 Fr Foley.
3. Lower midline incision; split recti; incise transversalis fascia.
4. Sweep peritoneum off the dome and open a vertical cystotomy.
5. Identify the orifices (ureteral catheter if concerned) and excise the sac.
6. Place a 20–22 Fr Foley, close in two layers, leak-test, and drain.

Step	Intravesical	Combined intra-/extravesical
Delivery	Sac everted or dissected from the pseudocapsule from inside; mucosal incision about **0.5 cm** round the neck	A finger in the sac pushes it out; the neck is incised from outside
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if the inner and outer incisions do not meet
Closure	Two layers 2-0 polyglactin with a secure seromuscular layer, which prevents recurrence	

Laparoscopic / robotic

- Use steep Trendelenburg and cystoscopy first, with pneumoperitoneum at **15 mm Hg**. Place the assistant port opposite the sac, circumscribe the neck, and excise and bag the sac with the ureter in view; staplers risk stones forming on the staples.
- If the sac sits at the ureteral hiatus, amputate the neck, make a new hiatus and reimplant over a double-J. Extraperitoneal, single-site and single-port variants exist.

Complications

- **Urinary extravasation is commonest**. It shows as high drain output, infected urinoma, or ileus and urinary ascites after transperitoneal surgery. Check drain creatinine and do CT cystography.

ALGORITHM

1. **Managing a leak**
2. If asymptomatic, keep Foley and drain.
3. If a symptomatic collection forms, drain it percutaneously.
4. Do a cystogram before removal; if the leak persists, choose nephrostomies or exploration and closure.

- Other complications are UTI, fistula, bladder spasm, wound infection and bowel injury.

- Unrecognised ureteric injury presents with acute kidney injury or flank pain; try a retrograde stent, and if it fails, place a nephrostomy and repair later.

> **KEY POINT**
> Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely, and fix the outlet first.

3 · BLADDER STONES, FOREIGN BODIES AND FISTULAE

A bladder stone signals stasis, so treat the cause too.

- The cause is **stasis**: bladder outlet obstruction, neurogenic bladder, or augmented and substituted bladders. Foreign bodies are mostly iatrogenic (mesh, migrated intrauterine device).
- **Non-contrast CT** is best for stone burden, prostate volume and hydronephrosis. Cystoscopy is needed only for haematuria, suspected cancer or prostate assessment.
- Treat the stone **and** its cause; the concomitant operation often picks the route (transvesical prostatectomy, diverticulectomy or enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation risks urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technically demanding

- **Percutaneous** access two fingerbreadths above the symphysis is dilated to a **30 Fr** sheath for a 24–26 Fr nephroscope. Stones under 1 cm are grasped whole and larger ones fragmented; the suprapubic catheter is clamped at 24–48 h. After prior surgery, image first to exclude interposed bowel.
- **Transvesical laparoscopy** uses a pneumovesicum of **8–12 mm Hg**, avoids the peritoneum and suits eroded mesh. Transperitoneal laparoscopic or robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy.
- **Open** cystolithotomy opens the bladder vertically, removes the stones, inspects for tumour and intravesical lobe, and closes in two layers with a drain.
- Refractory fistulae (colovesical, vesicovaginal) may need partial cystectomy, and a refractory urinary fistula may justify simple cystectomy.

4 · PARTIAL CYSTECTOMY FOR BENIGN DISEASE AND URACHAL SURGERY

- Indications are leiomyoma, paraganglioma, endometriosis, urachal anomalies and fistulae; most benign tumours need only transurethral resection (TUR).
- **Avoid it with small capacity or severe dysfunctional voiding.** Exclude malignancy first, because cancer needs wide margins and lymphadenectomy. Lesions near the orifice need reimplanting.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal ultrasound (US), MRI	Small endovesical: TUR; large or intramural: wide local resection

Lesion	Clue	Investigation	Treatment / trap
Paraganglioma	Episodic hypertension, hypotension or syncope on **micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG scan; yellow raised mucosa	**Alpha-blockade before any instrumentation**; partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (bright on T1); cystoscopy limited	Partial cystectomy (disease is extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture and antibiotics, with or without drainage; if recurrent, excise urachus with a dome cuff

- Dome and anterior lesions are approached extraperitoneally, posterior ones transperitoneally, with no node dissection. **Aspirate before opening to avoid spillage**.
- Excise full thickness with about a **2-cm margin** if capacity allows, with frozen-section margins, and reimplant if the orifice is involved.
- **Urachal excision** takes the whole tract with peritoneal wings and a small dome cuff, with **no wide margin or frozen section**; the umbilicus may be kept if benign. Open excision uses a periumbilical midline incision down to a transmural dome cuff.
- Complications are extravasation (commonest), ureteric injury and reduced capacity.

> **KEY POINT**
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur if incompletely excised, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, a margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · SIMPLE (SUPRATRIGONAL) CYSTECTOMY

A last resort that helps only if the bladder is the source.

- It treats end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, and interstitial cystitis/bladder pain syndrome (IC/BPS).
- Prove the bladder is the source, since non-vesical pain persists. In IC/BPS, success is predicted by a **small fibrotic bladder, small capacity under anaesthesia and Hunner lesions**.
- Diversion alone risks pyocystis, bleeding, pain and secondary cancer.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed
Bladder opened	Often, to find the trigone and bladder neck	Avoided
Excision	Bivalve and amputate near the trigone; stump oversewn and residual mucosa cauterised	En bloc

Steps are to divide the ureters, bivalve the bladder, divide the pedicles, amputate each half and cover the stump with posterior peritoneum. An ileal conduit usually follows, and robotic surgery may reduce complications. Drains stay 5–7 days, until drain creatinine equals serum, and complications are common.

6 · URETERAL REIMPLANTATION IN ADULTS

Adults need reimplantation mainly for injury and stricture.

- Indications are distal ureteral trauma, iatrogenic injury, **stricture** and late-presenting injury, and reimplantation alongside diverticulectomy or partial cystectomy. Primary VUR is rare.

- In thermal or gunshot injury, debride even normal-looking ureter; assess partial injury with on-table retrograde pyelogram or methylene blue.
- Strictures follow impacted stone, radiation or chronic inflammation. An atrophic kidney needs decompression, then a renogram.
- Measure the gap with antegrade and retrograde imaging and exclude malignancy. A **poorly functioning kidney needs nephrectomy, not reimplantation**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
> Refluxing reimplantation is acceptable in adults. Refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less; reaches the posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over a Kelly clamp	Clamp pulls ureter in

- Divide the ureter proximal to disease, with a frozen-section margin in stricture. Mobilise the bladder off peritoneum, dividing the ipsilateral medial umbilical ligament. Test reach on the **decompressed** dome.
- **Lich-Gregoir**: a small cystotomy forms the neo-orifice, detrusor is incised toward the bladder neck sparing mucosa, a spatulated 4-0 anastomosis is stented, and detrusor is closed over the ureter.
- **Politano-Leadbetter**: through an anterior cystotomy, a neo-orifice about 1.5 cm superior-medial to the native one leads into a submucosal tunnel.
- For a poorly vascularised ureter, use a non-transecting side-to-side refluxing anastomosis.
- Preserve periureteric adventitia, spatulate, and keep knots outside the lumen. The stent is a **6 Fr × 26–28 cm double-J**.
- Primary reimplantation suits defects up to **4 cm**; longer gaps need a psoas hitch or Boari flap, and renal descent adds up to **5 cm**. Further reach comes from dividing the contralateral medial umbilical ligament and pedicle.
- Enter **superior-medially, toward the bladder neck**, because lateral sites kink on filling.
- Open surgery is preferred for trauma; laparoscopic and robotic surgery give less blood loss and shorter stay with equal patency.
- Aftercare is a Foley for 1 week with a cystogram and a stent for **≈6 weeks**. A **leak** is managed by draining the urinoma and keeping the Foley and stent; if refractory, place a nephrostomy.
- **Late obstruction** usually follows stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site); short strictures are treated endoscopically and high-grade ones by redo.

> **KEY POINT**
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other.

7 · BRIDGING DISTAL URETERAL DEFECTS

P3 *Boari flap — reconstructive ladder, Boari dimensions (base ≥4 cm, apex ≥3 cm, ≤3:1, measured empty, reach up to 15 cm), psoas hitch nerves, contraindications, advancement flap*

Not in the paper answer:

- Nonrefluxing Boari length is the defect plus about 3 cm; longer flaps need a wider base.
- Beyond reach, options are transureteroureterostomy, ileal ureter and autotransplantation.

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains about 5 cm	Sutures superficial, parallel to the fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥**4 cm** , apex ≥**3 cm** , length:base ≤**3:1**

- Hitch plus flap shortens the flap and preserves capacity. The hitch sutures go into the psoas minor tendon, avoiding the **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- The Boari flap is based on the diseased side, anchored to psoas and tubularised in two layers.
- A bladder advancement flap gives comparable reach and is simpler. A **small-capacity, poorly compliant bladder** contraindicates both hitch and flap.
- Complications are femoral or genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of a narrow-based flap, and reduced capacity.

> **KEY POINT**
> The bladder is the donor tissue: hitch first, flap second. Every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · ENTEROCYSTOPLASTY (AUGMENTATION)

Augmentation protects the kidneys from high bladder pressure.

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation, especially with upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised**, which abolishes peristalsis.
- Work-up includes video-urodynamics, serum chemistry, B12 and bowel history; **confirm the patient can catheterise**. Metabolic effects: See Ch 133.
- Add a sling, artificial urinary sphincter, bladder neck closure, catheterisable stoma or nonrefluxing reimplant as needed.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to the ileocaecal valve (spares B12 and bile-salt absorption); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first
Stomach	Possible; needs a higher incision

> **ALGORITHM**
> 1. Isolate ileum with a broad mesenteric base and check reach to the dome.
> 2. Restore bowel side-to-side; close the window.
> 3. **Detubularise** on the antimesenteric border and fold into a U.
> 4. **Bivalve the bladder widely** and make a wide-mouthed anastomosis.
> 5. Suprapubic tube with or without Foley; leak-test; drain.

- A narrow mouth creates a poorly emptying pseudodiverticulum.

- Early complications are extravasation (preserve the arcade), UTI, bowel injury, fistula and intra-abdominal infection.
- **Perforation** follows missed catheterisation and is often masked by sensory loss. Diagnose with CT cystogram and proceed to **emergency exploration**.
- Stones are the commonest long-term complication. Neoplasia risk means surveillance endoscopy from **10 years**, with routine metabolic and vitamin bloods.
- If the urethra is not catheterisable, do a continent ileocaecocystoplasty; if the patient cannot catheterise, an incontinent ileovesicostomy.

KEY POINT

Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem. Detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance are the price.

9 · CONTINENT CATHETERISABLE CHANNELS

P2 *Principles of continence in continent diversion; Mitrofanoff principle — submucosal tunnel flap valve, appendix first choice, Yang-Monti alternative, channel catheter 2–4 weeks, continence >95%, stomal stenosis commonest*

Not in the paper answer:

- Channels suit adequate-capacity bladders after failed urethral CIC; augment if compliance is poor.
- The bladder is drained for 2–4 weeks before channel catheterisation starts. Always have a fallback if the appendix is absent or short.
- A Monti uses 2–4 cm of ileum; a double Monti risks channel diverticula; vas, colon and stomach channels do poorly.

ALGORITHM

1. **Minimally invasive appendicovesicostomy**
2. Detach the appendix with a caecal cuff and spatulate it.
3. Make a ≈4-cm submucosal detrusor tunnel and anastomose.
4. Bring the stoma to the right lower quadrant or umbilicus.
5. Drain the bladder for **2–4 weeks**. Continence is **>95%**.

KEY POINT

Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor, and missed catheterisation endangers the bladder, not just the channel.

10 · VESICOVAGINAL FISTULA (VVF) †

P3 P2 *VVF: management options, early vs delayed repair — causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps*

Not in the paper answer:

- Leakage usually starts 1–3 weeks after hysterectomy, and a coexisting ureteric injury is present in about 10%.
- A ureterovaginal fistula is treated with a ureteric stent first, and by ureteroneocystostomy if that fails.

★ Putting it together

A · NMIBC: RISK AND TREATMENT

AUA/SUO risk is assigned after complete TURBT.

Risk	Features	Adjuvant	Cystoscopy
Low	Low-grade (LG) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; high-grade (HG) Ta ≤3 cm	Induction MMC/gemcitabine or BCG; maintain 1 yr	3–6-monthly for 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; LVI; HG prostatic urethra	Re-TURBT (T1); BCG with **maintenance 3 yr** (SWOG); early RC if highest-risk	3–4-monthly for 2 yr, 6-monthly for 2 yr, then yearly; upper tracts yearly for 2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; if unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: CYSTECTOMY VS PRESERVATION SEE CH 131, 132

	NAC + RC + pelvic node dissection	Trimodality therapy
Who	Default if fit; NAC (gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, or NYHA III–IV: local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ after NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · CHOOSING THE DIVERSION SEE CH 133–135

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise	Default continent option, either sex	Unusable urethra or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40** ; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb
Main problems	Stoma problems, ureteroileal stricture	Night leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · METABOLIC COMPLICATIONS BY SEGMENT

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis**	Bicarbonate, potassium citrate
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	Sodium chloride; avoid jejunum
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria syndrome	H2 blocker/PPI

Losing terminal ileum lowers B12 and causes bile-salt diarrhoea; buffering acidosis in bone causes osteomalacia. See Ch 133

E · METASTATIC UROTHELIAL CARCINOMA

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: enfortumab vedotin (EV) with pembrolizumab (EV-302); otherwise gem–cis (gem–carbo if cisplatin-ineligible).
3. **If no progression on platinum**, give avelumab maintenance (JAVELIN Bladder 100).
4. **Later lines**: pembrolizumab; erdafitinib (FGFR-altered); EV; sacituzumab govitecan. See Ch 131